

1. Show that the product of two upper triangular matrices is an upper triangular matrix.

Solution: Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two upper triangular matrices, so that $a_{ij} = 0 = b_{ij}$ for all $j < i$. If $j < i$, then for any $k \in \{1, 2, \dots, n\}$, exactly one of $k \leq j < i$, $j < i \leq k$ and $j < k < i$ will be true. For the first case $a_{ik} = 0$, for the second case $b_{kj} = 0$, and for the third case $a_{ik} = 0 = b_{kj}$. Therefore, in each case, $a_{ik}b_{kj} = 0$. Therefore $\sum_{k=1}^n a_{ik}b_{kj} = 0$. Hence AB is an upper triangular matrix. $\square$

2. For an $n \times n$ matrix $A = [a_{ij}]$, the trace is defined as $\text{tr}(A) = a_{11} + \dots + a_{nn}$. Let A and B be two $n \times n$ matrices. If $AB = \mathbf{O}$ then show that $\text{tr}((A+B)^3) = \text{tr}(A^3) + \text{tr}(B^3)$.

Solution: As $AB = \mathbf{O}$, we have $(A+B)^3 = A^3 + BA^2 + B^2A + B^3$. We first prove that $\text{tr}(X+Y) = \text{tr}(X) + \text{tr}(Y)$ and $\text{tr}(XY) = \text{tr}(YX)$, where $X = [x_{ij}]$ and $Y = [y_{ij}]$ are two $n \times n$ matrices. It is easy to see that $\text{tr}(X+Y) = \sum_{i=1}^n (x_{ii} + y_{ii}) = \sum_{i=1}^n x_{ii} + \sum_{i=1}^n y_{ii} = \text{tr}(X) + \text{tr}(Y)$. Further,

$$\text{tr}(XY) = \sum_{i=1}^n \sum_{k=1}^n x_{ik}y_{ki} = \sum_{k=1}^n \sum_{i=1}^n x_{ik}y_{ki} = \sum_{k=1}^n \sum_{i=1}^n y_{ki}x_{ik} = \text{tr}(YX).$$

Therefore we have $\text{tr}(BA^2) = \text{tr}(A^2B) = \text{tr}(\mathbf{O}) = 0$. Similarly, $\text{tr}(B^2A) = 0$.

Hence $\text{tr}(A+B)^3 = \text{tr}(A^3) + \text{tr}(BA^2) + \text{tr}(B^2A) + \text{tr}(B^3) = \text{tr}(A^3) + \text{tr}(B^3)$. $\square$

3. Supply two examples for each of the following and explain their geometrical meaning.

- Two linear equations in two variables with exactly one solution.
- Two linear equations in two variables with infinitely many solutions.
- Two linear equations in two variables with no solutions.
- Three linear equations in two variables with exactly one solution.
- Three linear equations in two variables with no solutions.

Solution:

- Take $x + y = 2$, $x - y = 0$. They represent two lines in $\mathbb{R}^2$ intersecting at a point.
- Take $x + y = 2$, $2x + 2y = 4$. They represent the same line in $\mathbb{R}^2$.
- Take $x + y = 2$, $2x + 2y = 1$. They represent two parallel lines in $\mathbb{R}^2$, no intersection.
- Take $x + y = 2$, $x - y = 0$, $2x - y = 1$. They represent three lines in $\mathbb{R}^2$ with a single point in common.
- Take $x + y = 2$, $x - y = 0$, $2x + 2y = 1$. They represent three lines in $\mathbb{R}^2$ with no point in common.

Or, take $x + y = 2$, $x - y = 0$, $x + 2y = 1$. They represent three lines in $\mathbb{R}^2$ with no point in common but no two of them are parallel.

$\square$

4. For what values of $\lambda \in \mathbb{R}$, the following system of equations has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions?

$$(5 - \lambda)x + 4y + 2z = 4, \quad 4x + (5 - \lambda)y + 2z = 4, \quad 2x + 2y + (2 - \lambda)z = 2.$$

Also, find the solutions whenever they exist.

Solution: The augmented matrix of the given system of equations is

$$\begin{aligned} & \begin{bmatrix} 5 - \lambda & 4 & 2 & | & 4 \\ 4 & 5 - \lambda & 2 & | & 4 \\ 2 & 2 & 2 - \lambda & | & 2 \end{bmatrix} \\ & \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 2 & 2 - \lambda & | & 2 \\ 4 & 5 - \lambda & 2 & | & 4 \\ 5 - \lambda & 4 & 2 & | & 4 \end{bmatrix} \\ & \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 2 & 2 & 2 - \lambda & | & 2 \\ 0 & 1 - \lambda & -2 + 2\lambda & | & 0 \\ 5 - \lambda & 4 & 2 & | & 4 \end{bmatrix} \\ & \xrightarrow{R_3 \rightarrow R_3 - \frac{1}{2}(5 - \lambda)R_1} \begin{bmatrix} 2 & 2 & 2 - \lambda & | & 2 \\ 0 & 1 - \lambda & -2 + 2\lambda & | & 0 \\ 0 & \lambda - 1 & 2 - \frac{1}{2}(2 - \lambda)(5 - \lambda) & | & \lambda - 1 \end{bmatrix} \\ & \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 2 & 2 & 2 - \lambda & | & 2 \\ 0 & 1 - \lambda & -2 + 2\lambda & | & 0 \\ 0 & 0 & -\frac{1}{2}(\lambda^2 - 11\lambda + 10) & | & \lambda - 1 \end{bmatrix}. \end{aligned}$$

- (i) **No Solution:** In this case, the rank of the coefficient matrix $\neq$ the rank of the augmented matrix. Thus $\lambda^2 - 11\lambda + 10 = 0$ and $\lambda - 1 \neq 0$. That is, if $\lambda = 10$ then the given system has no solution.
- (ii) **Unique Solution:** In this case, the rank of the coefficient matrix = the rank of the augmented matrix = 3. Thus $\lambda^2 - 11\lambda + 10 \neq 0$. That is, if $\lambda \neq 1, 10$ then the given system has a solution. The unique solution is given by $\left[\frac{4}{10-\lambda}, \frac{4}{10-\lambda}, \frac{2}{10-\lambda}\right]$.
- (iii) **Infinitely Many Solutions:** In this case, the rank of the coefficient matrix = the rank of the augmented matrix < 3 . In this case, $\lambda^2 - 11\lambda + 10 = 0$ and $\lambda - 1 = 0$, that is, $\lambda = 1$. Thus, if $\lambda = 1$ then the given system has infinitely many solutions. The solution set is given by

$$\left\{ \left[1 - \alpha - \frac{\beta}{2}, \alpha, \beta \right]^t : \alpha, \beta \in \mathbb{C} \right\}.$$

□

5. Prove that the interchange of two rows of a matrix can be accomplished by a finite sequence of elementary row operations of the other two types.

Solution: Let A be an $m \times n$ matrix, where $m \geq 2$. Let the i -th and j -th rows of A be $\mathbf{u}$ and $\mathbf{v}$, respectively.

Let application of $R_j \rightarrow R_j + R_i$ on A transform A into A_1 , application of $R_i \rightarrow R_i + (-1)R_j$ on A_1 transform A_1 into A_2 , application of $R_j \rightarrow R_j + R_i$ on A_2 transform A_2 into A_3 , and application of $R_i \rightarrow (-1)R_i$ on A_3 transform A_3 into A_4 . Then the i -th and j -th rows of A_1 are $\mathbf{u}$ and $\mathbf{u} + \mathbf{v}$, respectively. The i -th and j -th rows of A_2 are $-\mathbf{v}$ and $\mathbf{u} + \mathbf{v}$, respectively. The i -th and j -th rows of A_3 are $-\mathbf{v}$ and $\mathbf{u}$, respectively. Finally, the i -th and j -th rows of A_4 are $\mathbf{v}$ and $\mathbf{u}$, respectively.

Thus, we see that the i -th and the j -th row of a matrix A can be interchanged by applying the sequence $R_j \rightarrow R_j + R_i$, $R_i \rightarrow R_i + (-1)R_j$, $R_j \rightarrow R_j + R_i$ and $R_i \rightarrow (-1)R_i$ of elementary row operations on A . $\square$

6. Let A be an $n \times n$ matrix. If the system $A^2\mathbf{x} = \mathbf{0}$ has a non-trivial solution then show that the system $A\mathbf{x} = \mathbf{0}$ also has a non-trivial solution.

Solution: Assume that the system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. Then for $\mathbf{y} \in \mathbb{R}^n$, we have

$$\begin{aligned} A^2\mathbf{y} = \mathbf{0} &\Rightarrow A(A\mathbf{y}) = \mathbf{0} \\ &\Rightarrow A\mathbf{y} = \mathbf{0}, \text{ since the system } A\mathbf{x} = \mathbf{0} \text{ has only the trivial solution} \\ &\Rightarrow \mathbf{y} = \mathbf{0}, \text{ since the system } A\mathbf{x} = \mathbf{0} \text{ has only the trivial solution} \\ &\Rightarrow \text{the system } A^2\mathbf{x} = \mathbf{0} \text{ has no non-trivial solution.} \end{aligned}$$

Aliter: Let $\mathbf{y}$ be a non-trivial solution of $A^2\mathbf{x} = \mathbf{0}$. Then we have $A^2\mathbf{y} = \mathbf{0}$. Now if $A\mathbf{y} = \mathbf{0}$ then $\mathbf{y}$ is a non-trivial solution of $A\mathbf{x} = \mathbf{0}$. If $A\mathbf{y} \neq \mathbf{0}$ then $A\mathbf{y}$ is a non-trivial solution of $A\mathbf{x} = \mathbf{0}$. $\square$

7. Let A and B be two 2×3 matrices that are in reduced row echelon form. If the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set then show that $A = B$.

Solution: There are only 7 possible 2×3 matrices that are in reduced row echelon form. These are

$$\begin{aligned} A_1 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & a & b \\ 0 & 0 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 1 & a \\ 0 & 0 & 0 \end{bmatrix}, \quad A_4 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \\ A_5 &= \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \end{bmatrix}, \quad A_6 = \begin{bmatrix} 1 & a & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad A_7 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ where } a, b \in \mathbb{R}. \end{aligned}$$

Let $\mathbf{x} = [x, y, z]^t \in \mathbb{R}^3$. Then we have

$$\begin{aligned} A_1\mathbf{x} = \mathbf{0} &\Rightarrow x, y \text{ and } z \text{ are arbitrary;} \\ A_2\mathbf{x} = \mathbf{0} &\Rightarrow x = -ay - bz, \text{ where } y \text{ and } z \text{ are arbitrary;} \\ A_3\mathbf{x} = \mathbf{0} &\Rightarrow y = -az, \text{ where } x \text{ and } z \text{ are arbitrary;} \\ A_4\mathbf{x} = \mathbf{0} &\Rightarrow z = 0, \text{ where } x \text{ and } y \text{ are arbitrary;} \\ A_5\mathbf{x} = \mathbf{0} &\Rightarrow x = -az, y = -bz, \text{ where } z \text{ is arbitrary;} \\ A_6\mathbf{x} = \mathbf{0} &\Rightarrow x = -ay, z = 0, \text{ where } y \text{ is arbitrary;} \text{ and} \\ A_7\mathbf{x} = \mathbf{0} &\Rightarrow y = z = 0, \text{ where } x \text{ is arbitrary.} \end{aligned}$$

We see that the above seven system of equations have seven distinct solution sets. Hence none of the above seven system of equations are equivalent. So, if A and B are two 2×3 matrices

that are in reduced row echelon form and if the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have same solution sets, then A must be equal to B . $\square$

8. Let A and B be two $m \times n$ matrices that are in reduced row echelon form. If the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set then show that $A = B$.

Solution: We will apply induction on the number of columns to prove that “if the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set and A, B are RREF of the same size then $A = B$ ”.

Base Case: Let both A and B have 1 column. Since there are only two RREF of an $m \times 1$ matrix, viz., $\mathbf{0}_{m \times 1}$ and e_1 , we find easily that if the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set then $A = B$.

Induction Hypothesis: Assume that for any two RREF P, Q of size $m \times k$, if the systems $P\mathbf{x} = \mathbf{0}$ and $Q\mathbf{x} = \mathbf{0}$ have the same solution set then $P = Q$.

Inductive Case: Let A and B be two RREF of size $m \times (k + 1)$. Let the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set. We want to show that $A = B$.

Let $\hat{A}$ and $\hat{B}$ be the matrices obtained by deleting the last columns of A and B , respectively. It is clear that $\hat{A}$ and $\hat{B}$ are also RREF, but having k columns. Let $\mathbf{u} = [u_1, \dots, u_k]^t$. We have

$$\hat{A}\mathbf{u} = \mathbf{0} \Leftrightarrow A \begin{bmatrix} \mathbf{u} \\ 0 \end{bmatrix} = \mathbf{0} \Leftrightarrow B \begin{bmatrix} \mathbf{u} \\ 0 \end{bmatrix} = \mathbf{0} \Leftrightarrow \hat{B}\mathbf{u} = \mathbf{0}.$$

Thus the systems $\hat{A}\mathbf{x} = \mathbf{0}$ and $\hat{B}\mathbf{x} = \mathbf{0}$ have the same solution set. By induction hypothesis, we find that $\hat{A} = \hat{B}$.

Let, if possible, $A \neq B$. Let $\mathbf{v} = [v_1, \dots, v_k, v_{k+1}]^t$ be a solution of $A\mathbf{x} = \mathbf{0}$. Then we have $(A - B)\mathbf{v} = \mathbf{0}$ which gives that $v_{k+1} = 0$. Hence the $(k + 1)$ -th column of A and B are leading columns. Let the $(k + 1)$ -th column of A and B be e_r and e_s , respectively, where $r \neq s$. Then the number of non-zero rows of $\hat{A}$ and $\hat{B}$ will be $r - 1$ and $s - 1$, respectively. As $r - 1 \neq s - 1$, this gives that $\hat{A} \neq \hat{B}$, a contradiction

Hence by mathematical induction, we conclude that if the systems $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set and A, B are RREF of the same size then $A = B$.

Aliter

Suppose, if possible, $A \neq B$. Let k be the smallest positive integer such that the k -th columns $\mathbf{a}_k$ and $\mathbf{b}_k$ of A and B , respectively, differ. Let A_{k-1} (resp. B_{k-1}) be the matrix formed by the first $k - 1$ columns of A (resp. B). Then $A_{k-1} = B_{k-1}$.

Now, it cannot happen that **both** $\mathbf{a}_k$ and $\mathbf{b}_k$ are leading columns, otherwise the leading 1's on them are at the same positions, and consequently we will get $\mathbf{a}_k = \mathbf{b}_k$.

Next neither of $\mathbf{a}_k$ and $\mathbf{b}_k$ is a zero column. If $\mathbf{a}_k$ is a zero column, then $\mathbf{b}_k$ is not (since they differ). So, $\mathbf{e}_k = [0, \dots, 1, \dots, 0]^t$ is a solution of $A\mathbf{x} = \mathbf{0}$ but not of $B\mathbf{x} = \mathbf{0}$.

Suppose $\mathbf{a}_k$ is a non-zero, non-leading column and suppose $a_{rk} \neq b_{rk}$. Consider the vector $\mathbf{s} = [s_1, s_2, \dots, s_k, 0, \dots, 0]^t$, where

$$s_i = \begin{cases} -a_{pk}, & \text{if the } i\text{-th column of } A \text{ is } \mathbf{e}_p \text{ (leading), } i < k; \\ 0, & \text{if the } i\text{-th column of } A \text{ is non-leading, } i < k; \\ 1, & \text{if } i = k. \end{cases}$$

We will now show that $As = \mathbf{0}$ but $Bs \neq \mathbf{0}$. Let $\mathbf{a}_k = [a_{1k}, \dots, a_{jk}, 0, \dots, 0]^t$, where $a_{1j} \neq 0$. Since $\mathbf{a}_k$ is a free column, we find that $[a_{j1}, \dots, a_{j,k-1}] \neq \mathbf{0}$ and $[a_{i1}, \dots, a_{i,k-1}] = \mathbf{0}$ for $i \geq j+1$. Thus, there are exactly j leading columns in A among the first k columns. Let these leading columns be $\mathbf{a}_{\alpha_1}, \dots, \mathbf{a}_{\alpha_j}$. Note that $\mathbf{a}_{\alpha_i} = \mathbf{e}_i$ for $1 \leq i \leq j$. Thus $As = \mathbf{0}$ iff

$$\begin{aligned} a_{1\alpha_1}s_{\alpha_1} + \dots + a_{1\alpha_j}s_{\alpha_j} + a_{1k} &= 0; \\ a_{2\alpha_1}s_{\alpha_1} + \dots + a_{2\alpha_j}s_{\alpha_j} + a_{2k} &= 0; \\ \vdots & \quad \quad \quad \vdots \quad \quad \quad \vdots \\ a_{j\alpha_1}s_{\alpha_1} + \dots + a_{j\alpha_j}s_{\alpha_j} + a_{jk} &= 0. \end{aligned}$$

Since $\mathbf{a}_{\alpha_i} = \mathbf{e}_i$ for $1 \leq i \leq j$, these equations give

$$\begin{aligned} s_{\alpha_1} + a_{1k} &= 0; \\ s_{\alpha_2} + a_{2k} &= 0; \\ \vdots & \quad \quad \quad \vdots \\ s_{\alpha_j} + a_{jk} &= 0. \end{aligned}$$

Now from the definition of $\mathbf{s}$, it is clear that these equations are satisfied, that is, $As = \mathbf{0}$. However, the r -th entry of Bs is $s_{\alpha_r} + b_{rk}$. As $s_{\alpha_r} = -a_{rk}$ and $a_{rk} \neq b_{rk}$, we find that the r -th equation of $B\mathbf{x} = \mathbf{0}$ is not satisfied by $\mathbf{s}$.

Thus $\mathbf{s}$ is a solution of $A\mathbf{x} = \mathbf{0}$ but not of $B\mathbf{x} = \mathbf{0}$. This is a contradiction. $\square$

9. Prove or disprove: There exist two solutions $\mathbf{x}_1$ and $\mathbf{x}_2$ of some consistent non-homogeneous system $A\mathbf{x} = \mathbf{b}$ such that $\mathbf{x}_1 + \mathbf{x}_2$ is also a solution of $A\mathbf{x} = \mathbf{b}$.

Solution: Consider a consistent non-homogeneous system $A\mathbf{x} = \mathbf{b}$. Here we have $\mathbf{b} \neq \mathbf{0}$. Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be two solutions of $A\mathbf{x} = \mathbf{b}$ so that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$. Then we have $A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{b} + \mathbf{b} \neq \mathbf{b}$, since $\mathbf{b} \neq \mathbf{0}$. Hence $\mathbf{x}_1 + \mathbf{x}_2$ can never be a solution of $A\mathbf{x} = \mathbf{b}$. Thus the given statement is disproved. $\square$

10. Prove or disprove: If two matrices of the same order have the same rank then they must be row equivalent.

Solution: Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. Here A and B have the same rank but they are not row equivalent (why?). Thus the given statement is disproved by this counterexample. $\square$